



**Assam Engineering
College**

*An Industrial IoT Monitoring System for Anomaly Detection Using
Machine Learning Algorithms*

Presented By:

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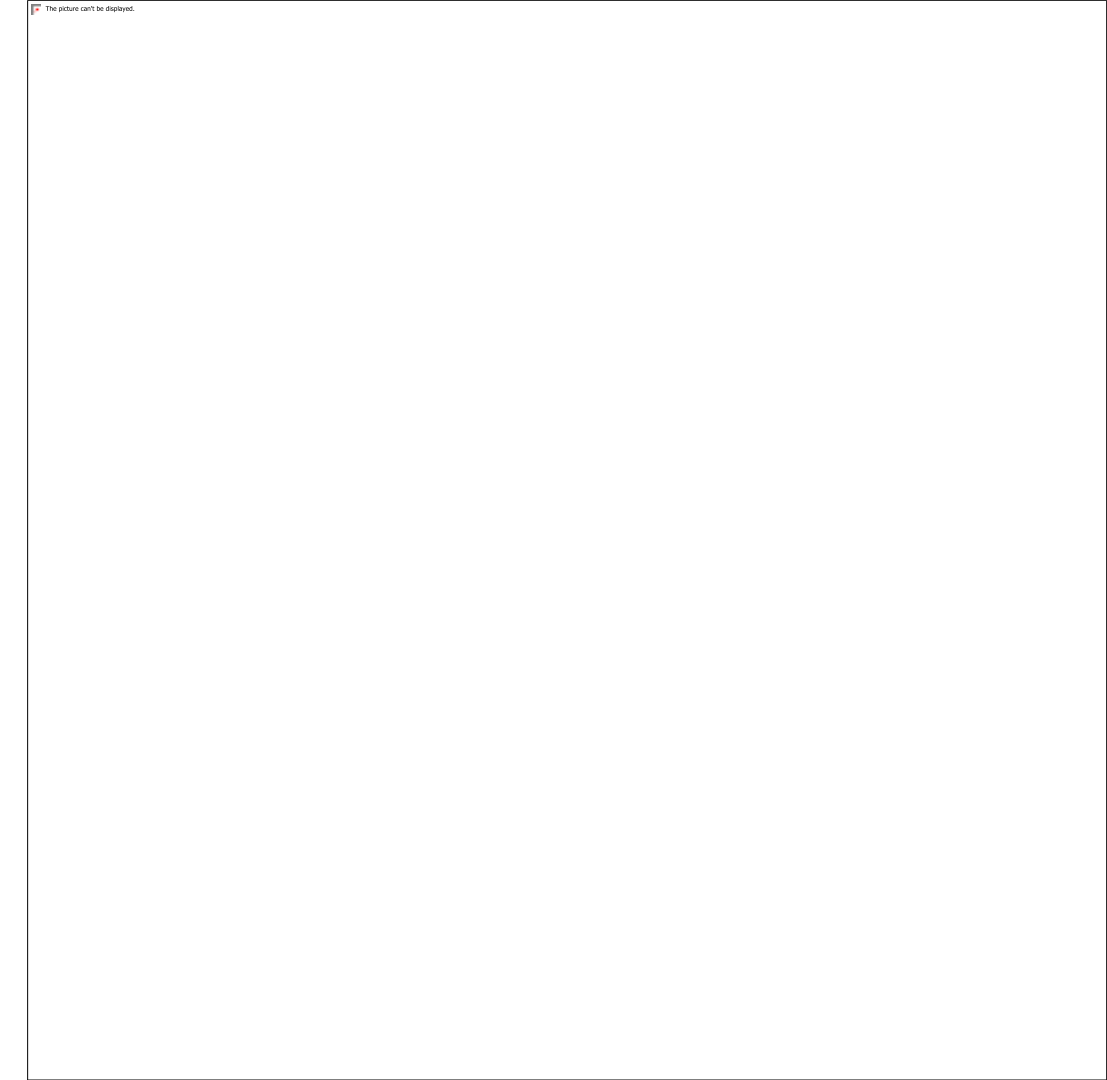
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Introduction & Motivation

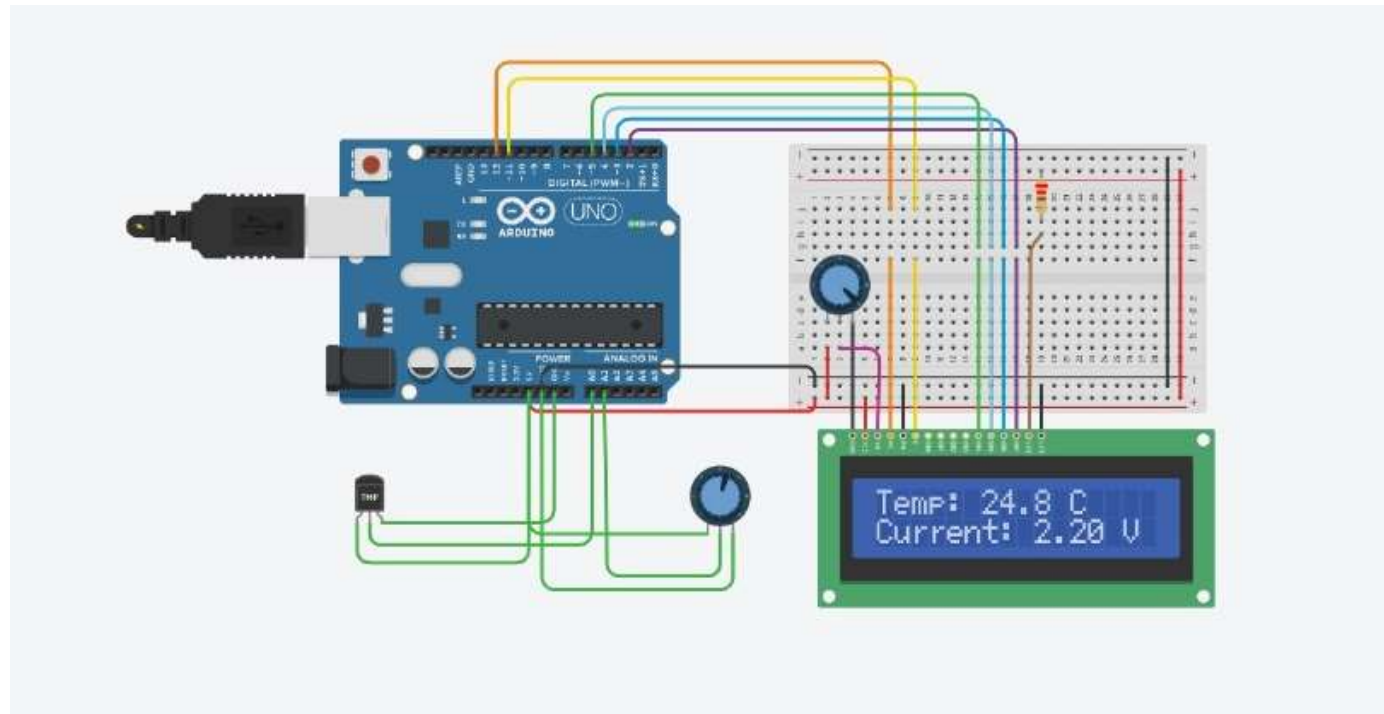
- ***The Cost of Downtime:*** Unexpected equipment failures cost industrial facilities up to 15% of annual revenue due to reactive maintenance strategies [1, 2].
- ***Limitations of Traditional SCADA:***
 - Heavily relies on static, rule-based thresholds.
 - Limits adaptability to seasonal shifts and equipment aging [3].
 - Often overlooks early-warning degradation signals, resulting in false alarms or undetected early-stage faults.

The Proposed Solution

- ***The Paradigm Shift:*** *Moving from calendar-driven to condition-based predictive maintenance by directly integrating Machine Learning into IIoT pipelines [17, 18, 19].*
- ***Our Contribution:*** *A streamlined, end-to-end IIoT platform that directly couples real-time sensory data collection with an active ML inference layer.*
- ***Key Features:***
 - *Evaluates 7 critical parameters simultaneously.*
 - *Utilizes sliding window feature extraction for continuous monitoring.*
 - *Maintains strict sub-200 ms inference latencies.*

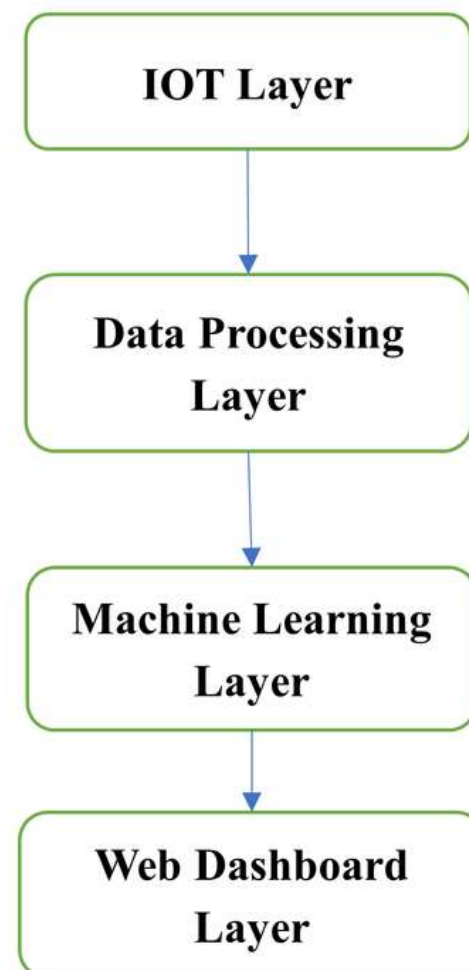


Hardware Setup & IoT Layer



- **Microcontroller:** Arduino Uno processes and packages values via JSON over USB serial.
- **Sensor Integration:**
 - **DHT11:** Temperature and humidity sampling (pin 2, 10 k Ω pull-up).
 - **ACS712:** Hall-effect based current measurement (analog pin A0).
- **Practical Application:** Demonstrates a highly practical, low-cost deployment approach for industrial monitoring.

High-Level System Architecture



- **Four Core Modules:**

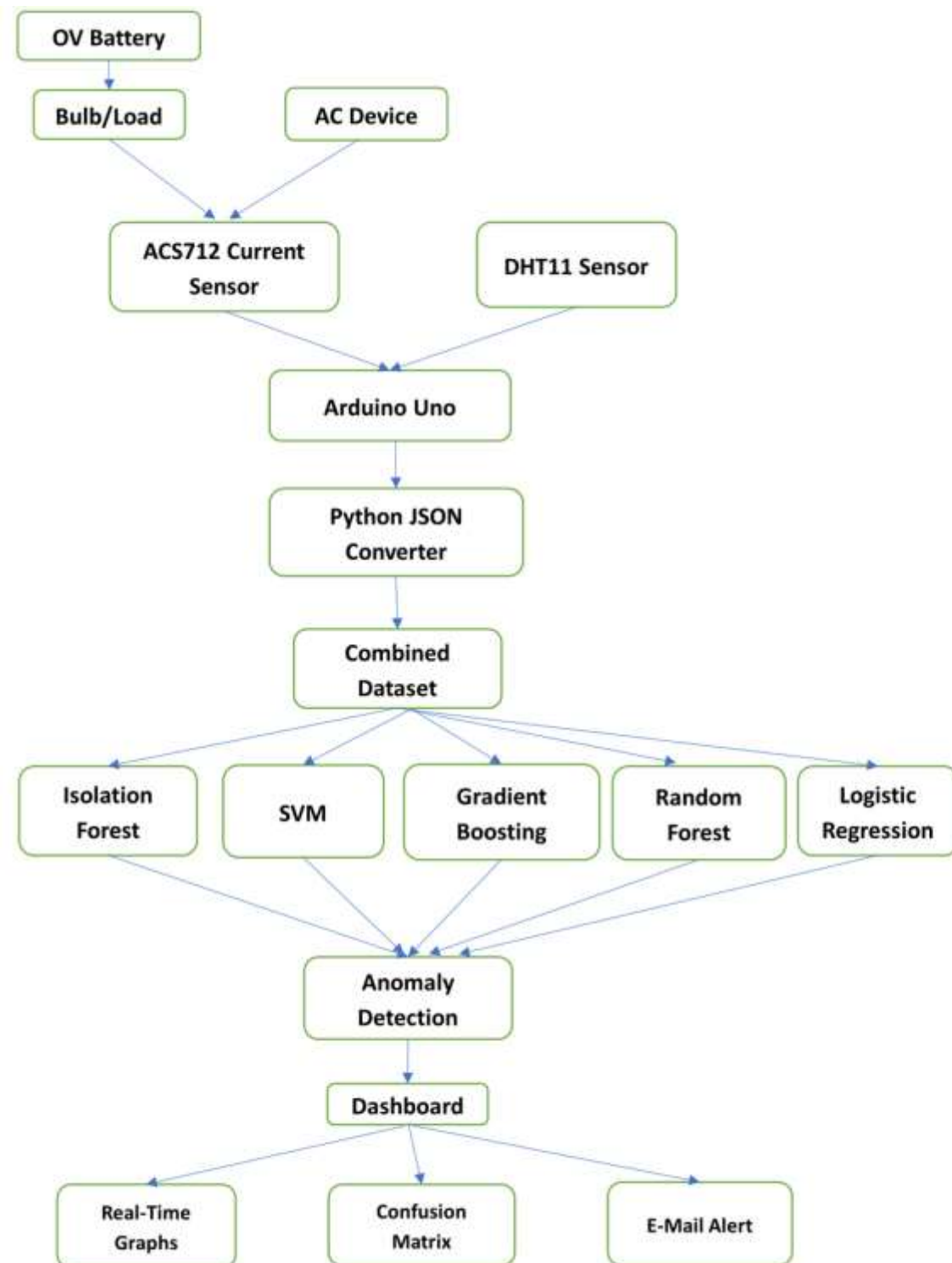
a. IoT Layer: Continuous physical environment sampling.

b. Data Processing: Cleaning, timestamping, and segmentation.

c. ML Inference: Feature evaluation and anomaly detection.

d. Web Dashboard: Real-time visualization for operators.

Hardware and Data Acquisition Setup



- *Arduino Uno integrated with DHT11 and ACS712 modules*
- *Continuous sampling and host transfer via USB serial (9600 baud)*
- *Low-cost, reproducible setup suitable for lab/industrial prototyping*
- *Forms the physical front-end for digital monitoring and analytics*

Methodology & Feature Engineering

- **Data Generation & Injection:**

- Generated 30 days of 1-minute telemetry (43,200 samples) across seven sensors.
- Controlled 5% synthetic anomaly injection (Gaussian deviations/spikes) to mimic realistic fault signatures.

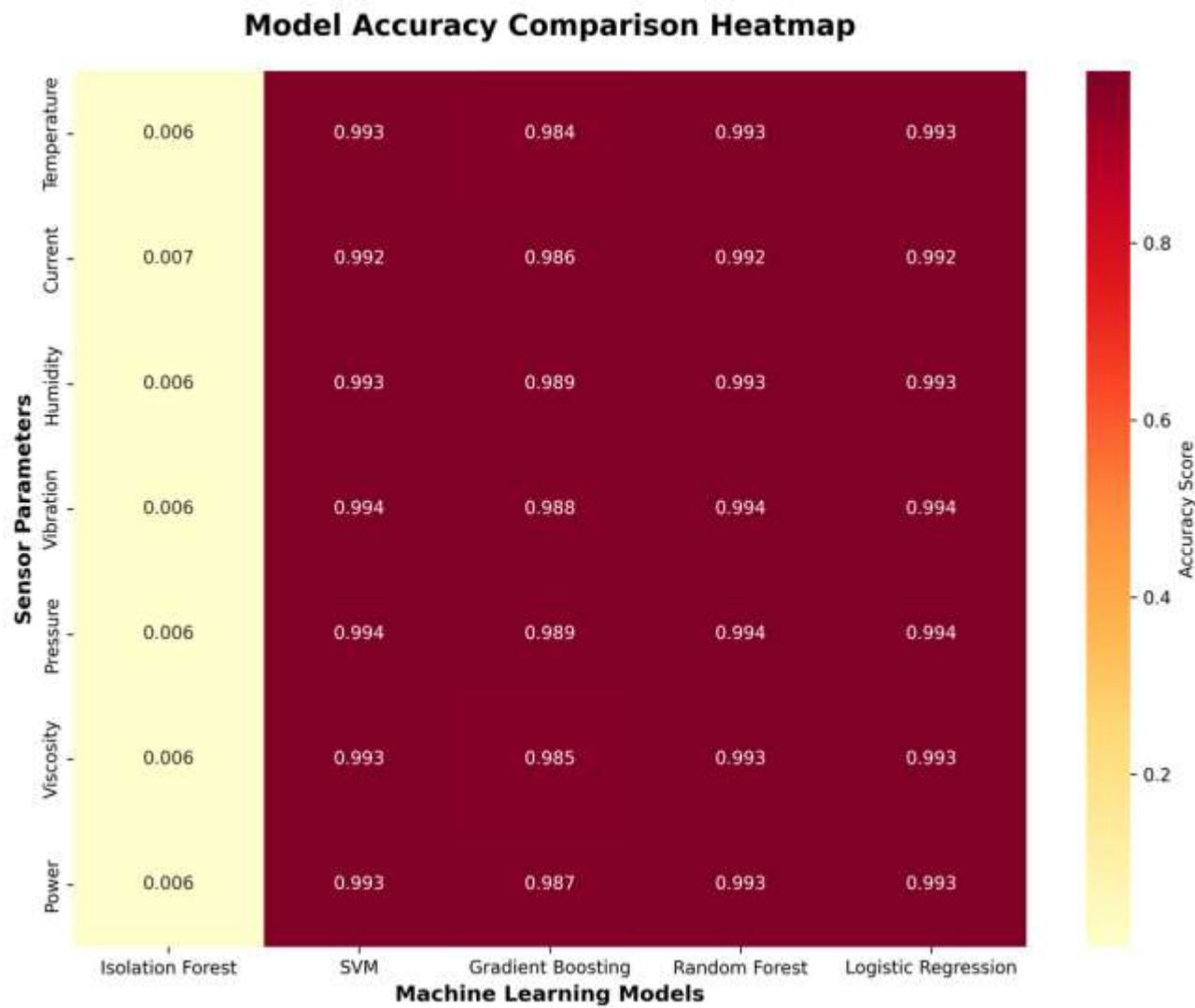
- **Signal Processing via Sliding Window [4]:**

- Continuous streams segmented into 24-step overlapping windows.
- Extracted dynamic statistical features (mean, standard deviation, skewness, kurtosis, etc.) [15].
- **Advantage:** Captures temporal patterns continuously for sub-second monitoring capabilities.

Evaluated Machine Learning Algorithms

- ***Evaluating diverse paradigms to reflect different modeling assumptions over mere hyperparameter optimization.***
- ***Unsupervised:*** Isolation Forest (Best for scarce labeled data, random feature space partitioning) [6].
- ***Kernel-Based:*** Support Vector Machine (RBF kernel for capturing non-linear relationships) [7].
- ***Ensemble (Tree-Based):***
 - Random Forest (Highly robust generalization) [9].
 - Gradient Boosting (Sequential error-correcting trees) [8].
- ***Linear/Probabilistic:*** Logistic Regression (Standard baseline model for speed and interpretability) [16].

Experimental Results (Synthetic Benchmark)



- **Top Performers:** SVM, Random Forest, and Logistic Regression achieved an average 99.35% accuracy across all parameters.
- **Class-Imbalance Adjustments:** Precision, Recall, and F1-Scores were evaluated due to the extreme dataset imbalance, with Logistic Regression yielding the highest recall.

Sensor Parameters	Isolation Forest	SVM	Gradient Boosting	Random Forest	Logistic Regression
Temperature	0.58%	99.32%	98.92%	99.32%	99.32%
Current	0.49%	99.47%	98.60%	99.47%	99.47%
Humidity	0.65%	99.25%	98.51%	99.25%	99.25%
Vibration	0.51%	99.44%	98.65%	99.44%	99.44%
Pressure	0.63%	99.33%	98.66%	99.33%	99.33%
Viscosity	0.61%	99.28%	98.85%	99.28%	99.28%
Power	0.56%	99.34%	98.90%	99.34%	99.34%
Average	0.58%	99.35%	98.73%	99.35%	99.35%

Real-World Dataset Validation

Model	Precision	Recall	F1-Score
Isolation Forest	0.0073	0.0548	0.0128
SVM	0.0072	0.5029	0.0142
Gradient Boosting	0.0065	0.2086	0.0126
Random Forest	0.0000	0.0000	0.0000
Logistic Regression	0.0074	0.5293	0.0147

- **Addressing Practical Challenges:** Validated on a real industrial water pump telemetry dataset.
- **Specs:** 44,807 samples, 50 sensor channels.
- **Unsupervised Performance:** Isolation Forest achieved 66.34% accuracy on heavily imbalanced, weakly-labeled industrial anomalies.
- **Takeaway:** Demonstrated successful true-to-life operability, while transparently identifying the domain-shift challenges inherent in adapting from synthetic to real-world noisy data.

Load Testing & Visualization Dashboard



- **Sustained Throughput:** 298.5 requests/second.
- **Inference Speed:** < 6 ms p99 latency for single-stream operation.
- **Dashboard Utility:** Operator interface utilizing live gauges, rolling time-series trends, and automated email alerts.

Conclusion & Future Scope

- **Conclusion:** *Coupling low-cost hardware with supervised algorithms successfully mitigates unexpected downtime via sub-200 ms latency predictions and $\sim 99.35\%$ accuracy under ideal conditions.*
- **Future Directions:**
 - *Mitigate domain-shift between synthetic and real-world conditions with adaptive features.*
 - *Scale highly-concurrent throughput (beyond 10 streams) via GPU-assisted inference.*
 - *Acquire richer fault-label sets for supervised models in reality.*

References

- **[1]** A. Mussomeli, S. Laaper, B. Sniderman, and C. Chandra, "Making maintenance smarter: Predictive maintenance and the digital supply network," *Deloitte Insights*, 2017.
- **[2]** Henkel Adhesive Technologies, "The Impact of Unplanned Downtime in Industrial Manufacturing," *Henkel Adhesives*, 2020.
- **[3]** E. H. E. Suryadarma and T. J. Ai, "Predictive maintenance in SCADA-based industries: A literature review," *International Journal of Industrial Engineering and Engineering Management*, vol. 2, pp. 1-12, 2020.
- **[4]** L. Kulanuwat et al., "Anomaly Detection Using a Sliding Window Technique and Data Imputation with Machine Learning for Hydrological Time Series," *Water*, vol. 13, p. 1862, 2021.
- **[6]** F. T. Liu, K. M. Ting, and Z. Zhou, "Isolation Forest," *Proc. 8th IEEE Int. Conf. Data Mining (ICDM)*, pp. 413-422, 2008.
- **[7]** C. Cortes and V. Vapnik, "Support-vector Networks," *Machine Learning*, vol. 20, pp. 273-297, 1995.
- **[8]** J. H. Friedman, "Greedy Function Approximation: A Gradient Boosting Machine," *Annals of Statistics*, vol. 29, pp. 1189-1232, 2001

References

- **[9]** L. Breiman, "Random Forests," *Machine Learning*, vol. 45, pp. 5–32, 2001.
- **[15]** C. Bergmeir and J. M. Benítez, "On the use of cross-validation for time series predictor evaluation," *Information Sciences*, vol. 191, pp. 192–213, 2012.
- **[16]** D. R. Cox, "The regression analysis of binary sequences," *Journal of the Royal Statistical Society: Series B*, vol. 20, pp. 215–232, 1958.
- **[17]** R. S. S. Nuvvula et al., "Machine Learning-Driven Predictive Maintenance Framework for Anomaly Detection and Prognostics in Wind Farm Operations," *IEEE Access*, 2024.
- **[18]** E. Omol, L. W. Mburu, and D. A. Onyango, "Anomaly detection in IoT sensor data using machine learning techniques for predictive maintenance in smart grids," *International Journal of Science, Technology & Management*, 2024.
- **[19]** M. F. Pekşen, U. Yurtsever, and Y. Uyaroğlu, "Enhancing electrical panel anomaly detection for predictive maintenance with machine learning and IoT," *Alexandria Engineering Journal*, 2024.